# Efficient Duplicate Question Detection with Bi-Encoder Compression and Quantized Cross-Encoders

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**Abstract**

We study duplicate question detection under a deployment-oriented evaluation regime that balances predictive quality with efficiency. Starting from traditional sparse-text baselines, we compare a Siamese LSTM, a compact Bi-Encoder, and a BERT Cross-Encoder. The main contribution is a consistent benchmarking protocol that measures accuracy, F1, log loss, inference latency, and model size under the same validation/test split and a uniform batch size of 512. We further compress the Bi-Encoder and Cross-Encoder with int8 quantization. The results show that the BERT Cross-Encoder achieves the strongest predictive performance, while the early-stopped Bi-Encoder provides the best accuracy-efficiency compromise. Int8 quantization reduces storage substantially for both encoder families, but the runtime effect differs: Bi-Encoder latency remains nearly unchanged, whereas the quantized BERT Cross-Encoder becomes slower due to embedding reconstruction and dynamic quantization overhead.

## 1 Introduction

Duplicate question detection is a practical semantic matching task that appears in search, customer support, and community moderation. The usual modeling choice is not simply the most accurate classifier, but the one that delivers acceptable quality under memory and latency constraints. This report evaluates that trade-off on a shared benchmark and focuses on three questions: (i) whether a compact Bi-Encoder is a viable alternative to a heavier Cross-Encoder, (ii) how much int8 quantization reduces model footprint, and (iii) whether compression improves or degrades inference speed.

## 2 Methods

### 2.1 Dataset

We use the Quora Question Pairs dataset, a binary duplicate-question benchmark in which each example contains two natural-language questions and a label indicating whether they are semantically equivalent. The cleaned benchmark used in this project contains 323,432 training examples, 40,429 validation examples, and 40,429 test examples. This task is a good testbed for model comparison because it mixes lexical overlap with paraphrase matching and requires both surface-form and semantic reasoning.

### 2.2 Model families

We compare three groups of models. First, classical baselines include TF-IDF logistic regression and linear SVM, which offer strong sparse-text performance with very low latency. Second, a Siamese

LSTM encodes each question independently and compares the resulting vectors. Third, a compact Bi-Encoder learns a shared representation with mean pooling and pairwise comparison features. For the transformer family, we use a BERT Cross-Encoder that jointly encodes the pair and yields the strongest semantic interaction modeling.

### 2.3 Quantization

Quantization is applied as a post-training compression step. For both encoder families, linear layers are dynamically quantized to int8. In addition, the token embedding tables are stored in int8 form and reconstructed at runtime. This choice is motivated by model-size reduction: for the compact Bi-Encoder, the embedding tables dominate storage; for the BERT Cross-Encoder, the embedding and linear weights account for a large fraction of the checkpoint size.

### 2.4 Evaluation protocol

All final comparisons are run on the same validation and test splits with a uniform batch size of 512. Classification thresholds are selected on the validation set by maximizing F1, breaking ties with accuracy. This avoids arbitrary thresholding and ensures that reported test metrics reflect the best operating point on the validation distribution.

## 3 Results

Table 1 summarizes the representative models. The best traditional baseline is the linear SVM, but its F1 remains far below the neural models. The compact Bi-Encoder materially improves F1 while staying lightweight, and the int8 version reduces model size to roughly one quarter of the float checkpoint with only a small F1 drop. The BERT Cross-Encoder remains the strongest model in absolute accuracy and F1, but its computational cost is substantially higher than the Bi-Encoder family.

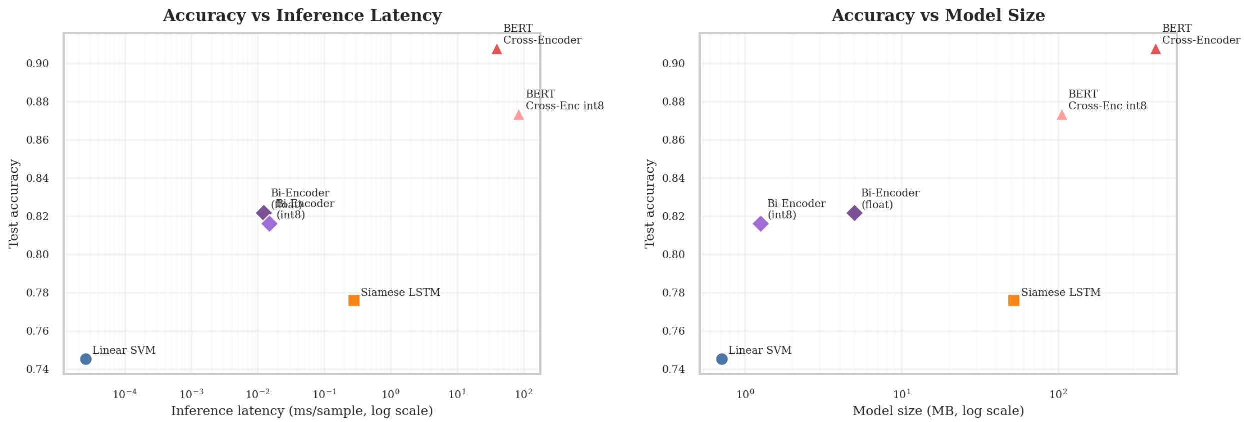
The best F1 in the full benchmark is achieved by BERT Cross-Encoder (0.8805), while the smallest shortlisted model is LR (Hand Features) (0.001 MB). The fastest shortlisted model is Linear SVM (TF-IDF) (0.000026 ms/sample).

These results follow the structure of the task. The linear SVM performs well because many duplicate questions share important words or phrases, so sparse TF-IDF features already capture a strong portion of the signal. The Siamese LSTM improves on simple linear models by learning dense sentence vectors, but it still compresses each question separately and therefore cannot directly compare tokens across the two questions. The Bi-Encoder is stronger because its learned representation is more expressive than the LSTM baseline, yet it stays compact enough to remain fast.

The BERT Cross-Encoder leads the benchmark because it models pairwise token interactions inside the transformer stack. That is particularly useful when duplicates are paraphrases with weak lexical overlap or when the same intent is expressed through different wording. In other words, the Cross-Encoder pays more computation to model richer cross-question alignment, and the benchmark shows that this extra capacity translates into the best F1 and accuracy.

### 3.1 BERT-specific comparison

Table 2 shows that BERT compression is storage-efficient but not necessarily latency-efficient. The float checkpoint reaches 0.9079 accuracy and 0.8805 F1 at 39.13 ms/sample. The int8 checkpoint




(a) Accuracy vs latency.

(b) Accuracy vs model size.

Figure 1: Publication-style trade-off plots for the representative models.

| Model | Acc. | F1 | Log loss | Latency (ms/sample) | Size (MB) |
|---|---|---|---|---|---|
| Linear SVM (TF-IDF) | 0.7455 | 0.6395 | 0.5918 | 0.000026 | 0.710 |
| Word2Vec + Siamese LSTM | 0.7759 | 0.7238 | 0.4646 | 0.281105 | 52.308 |
| Bi-Encoder Early Stopping | 0.8217 | 0.7747 | 0.5109 | 0.012353 | 4.982 |
| Embedding+Linear Int8 Bi-Encoder Early Stopping | 0.8163 | 0.7723 | 0.5193 | 0.014884 | 1.254 |
| BERT Cross-Encoder | 0.9079 | 0.8805 | 0.2199 | 39.132051 | 417.727 |
| BERT Cross-Encoder Int8 | 0.8736 | 0.8499 | 0.2712 | 84.541235 | 104.898 |

Table 1: Main benchmark summary on the full validation/test split with a uniform batch size of 512.

reduces the model size from 417.7 MB to 104.9 MB, but latency rises from 39.13 ms/sample to 84.54 ms/sample. This indicates that dynamic quantization plus int8 embedding reconstruction introduces runtime overhead that is visible on CPU.

The latency increase is not a contradiction. The int8 checkpoint stores weights more compactly, but the implementation reconstructs embeddings at runtime and still relies on dynamic quantization wrappers for the linear layers. That reduces memory footprint, yet it can add enough CPU overhead to offset the benefit in a full-dataset latency measurement. The takeaway is that compression and speed are related, but they are not the same objective.

# 4 Discussion

The benchmark reveals a clear hierarchy. If the objective is pure predictive performance, BERT Cross-Encoder is the best choice. If the objective is balanced deployment, the early-stopped Bi-Encoder is a better engineering trade-off because it maintains strong F1 while keeping the checkpoint small and the inference path fast. Quantization is most compelling for the Bi-Encoder, where storage drops sharply with almost no change in latency. For BERT, quantization does reduce storage, but the runtime behavior is less favorable because the compression strategy is not aligned with CPU inference efficiency.

The 6-epoch Bi-Encoder underperforms the early-stopped version, which suggests the model reaches its generalization peak before the longer schedule finishes and then begins to overfit or saturate. That makes the validation-selected checkpoint more reliable than simply training longer.

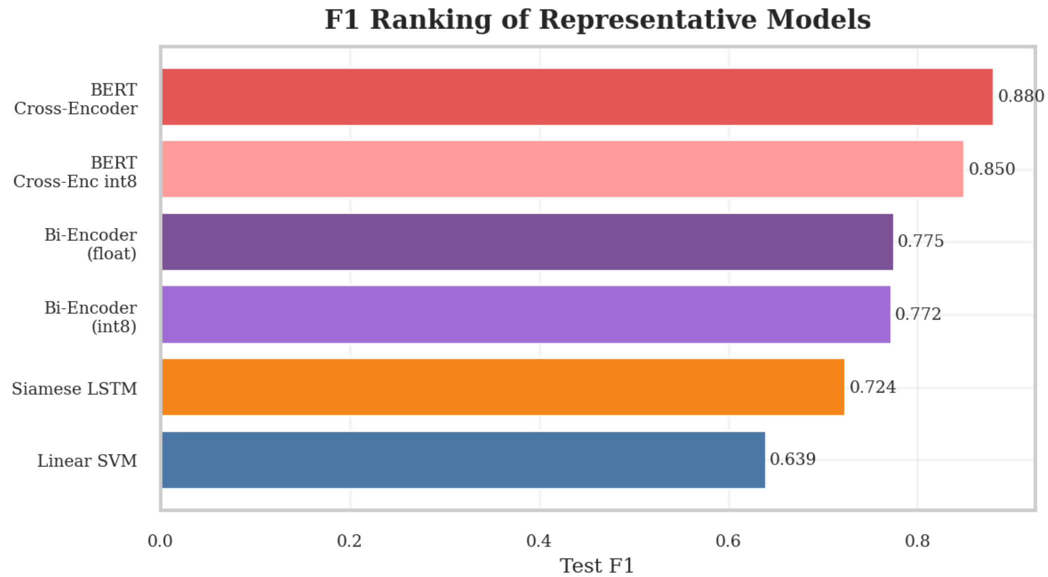



Figure 2: F1 ranking of representative models.

| Model | Acc. | F1 | Latency (ms/sample) | Size (MB) |
|---|---|---|---|---|
| BERT Cross-Encoder | 0.9079 | 0.8805 | 39.132051 | 417.727 |
| BERT Cross-Encoder Int8 | 0.8736 | 0.8499 | 84.541235 | 104.898 |

Table 2: Full-dataset comparison between the float and int8 BERT cross-encoder under the same batch size and evaluation split.

The quantized Bi-Encoder variants preserve most of the float model's quality, which indicates that most of the decision boundary is already captured in the learned representation and does not require full float precision.

The full benchmark also highlights a common mistake in model selection: storage and latency should not be assumed to move together. A smaller checkpoint can still be slower if the compression mechanism introduces extra reconstruction work at inference time. That is what happens for the BERT int8 variant here, whereas the Bi-Encoder benefits from quantization because its architecture is already lightweight and does not depend on a heavy pairwise attention stack.

## 5 Conclusion

We evaluated duplicate question detection under a unified, deployment-oriented benchmark. The main conclusion is that the best model depends on the objective. BERT Cross-Encoder delivers the highest accuracy and F1, but the compact Bi-Encoder offers the best balance of quality, speed, and memory footprint. Int8 quantization is highly effective for reducing storage, especially for the Bi-Encoder, but its runtime effect depends on the architecture and the implementation details of the inference path.

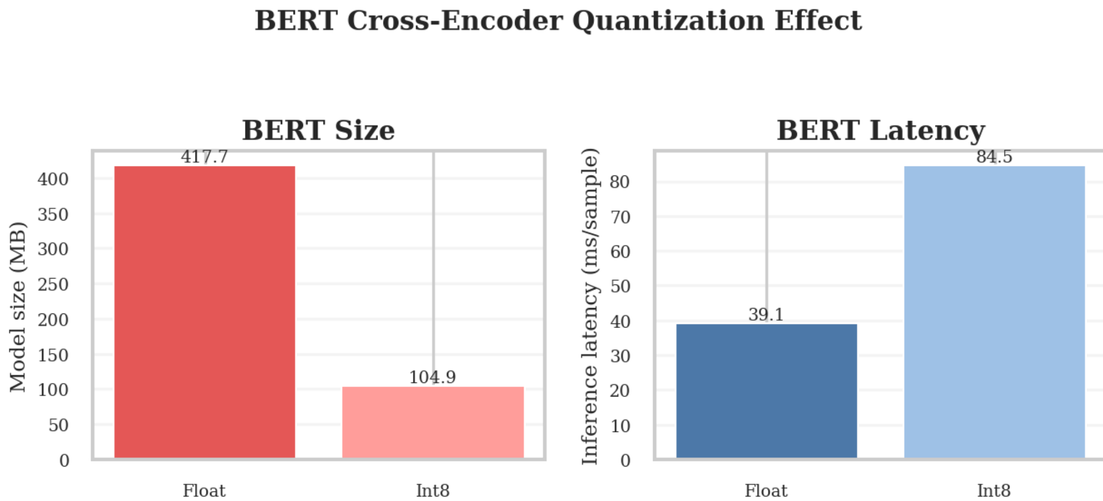



Figure 3: Effect of int8 quantization on the BERT Cross-Encoder.

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